

MY TAKE REPORT

DDOT TOPS Modernization, Demo Strategy & Requirements Brief

Subject:	TOPS 2.0 Modernization Demo, Strategy, Scope & Requirements for John Yzaguirre
Prepared by:	PMO Lead
Source document:	DDOT TOPS BPR Statement of Objectives, v2.0 (04/24/2023)
Purpose:	Win the TOPS Modernization contract by demonstrating a best-in-class, agentic-AI-powered permitting platform tailored to DDOT.

1. Overall Take

The demo needs to feel like a modernized DDOT permitting operation, not just an AI showcase.

The SOO describes a cloud-based TOPS 2.0 that improves efficiency, transparency, user experience, workflow automation, GIS/location intelligence, payments, reporting, security, accessibility, and coordination across applicants, reviewers, technicians, inspectors, and management. It is a top-level objectives document, not the final SOW/PWS, which means we have room to shape the vision, but the demo must be clearly positioned as aligned to the SOO rather than presented as a finished product definition.

Strategic Framing

The strongest approach is a thin-slice end-to-end demo that covers the actual permit lifecycle described in the SOO: intake, review, issuance, payment, status visibility, inspection/compliance, enforcement, renewals/extensions, and management reporting. The demo should signal that we understand DDOT's full modernization challenge, not just the front-end intake problem.

2. Recommended Primary Permit Journey

Center the demo on one permit type that is both map-heavy and workflow-heavy. The two strongest candidates are:

1. Construction/Excavation, high complexity, strong GIS story, frequent conflicts with utilities and other permits
2. Parking/Occupancy, high volume, clear public impact, relatable to both staff and applicants

Make it clear the platform supports all permit types listed in the SOO: Parking/Occupancy, Construction/Excavation, Commercial Vehicle, Public Space Rental/Annual, and Special/Heritage Tree Permits. Show the second permit type briefly in the type-selector screen.

3. Recommended Demo Flow (Screen by Screen)

The demo should walk through eight stages covering the complete permit lifecycle. Each stage maps directly to SOO objectives.

Stage 1, Login & Role Selection

Show the MFA prompt and a role switcher that lets the demo audience toggle between Applicant, Reviewer, and Admin/Manager views. RBAC should be immediately visible through what each user can and cannot see. The SOO specifies over a dozen distinct roles, the architecture should visually signal it supports the full roster, even if only three are demonstrated.

Stage 2, Applicant: Account & Project Setup

User creates or accesses an account, associates with a business account, and designates a project. This directly reflects SOO language on business accounts, project designation, and assigning/removing applicants to projects. This is often skipped in generic demos but is explicitly called out in the SOO as a requirement.

Stage 3, Applicant: Permit Application Intake

1. Permit type selector (show multiple types exist; focus on one)
2. GIS map for location selection with address auto-population from DC's Master Address Repository
3. Adaptive form fields that change based on permit type
4. Document and plan upload with electronic submission
5. Fee estimator visible before submission, the SOO requires transparent fee communication upfront
6. Pre-submission AI check: validates required fields, flags missing items, estimates review time, surfaces relevant DCMR regulation references (Titles 6, 9, 16, 18, 24)

SOO Note on Pre-Application Activity

The SOO's service delivery model explicitly starts with pre-application processes including Early Work Review and Non-permit inspections. Even a brief touchpoint here, a 'Pre-Application Consultation' banner or entry point, signals full lifecycle awareness.

Stage 4, Applicant: Submission & Status Dashboard

Shows submission receipt, a visible review clock/SLA countdown (the SOO is specific about configurable review timeframes that must be communicated to applicants and the public), real-time status, and notification preferences. This is one of the clearest themes in the SOO.

Stage 5, Reviewer: Work Queue & Electronic Plan Review

A DDOT reviewer sees routed applications, review clock warnings, uploaded plans with redline and comment tools, prior notes, and an AI-generated application summary. The reviewer can return the application, request more information, or advance to approval. This is one of the most explicitly described workflows in the SOO and should be one of the strongest screens in the demo.

Stage 6, GIS Conflict & Coordination View

Map view showing the permit location with overlaid nearby permits, utility work, DDOT projects, ADA barriers, and asset management data (sidewalk/street status). AI flags potential conflicts and surfaces

coordination recommendations. This is one of the most compelling DDOT-specific elements, the SOO specifically calls out geography-based coordination, conflict checking, and tying permits to block/ANC/Ward for compliance and reporting.

Asset Management Engine

The SOO explicitly calls for an asset management engine for permitting assets like sidewalks and streets. Surfacing this in the GIS layer, even as a labeled overlay, is a low-lift, high-signal differentiator that shows domain depth.

Stage 7, Applicant: Response & Resubmission

Applicant receives a notification, views reviewer comments, responds, and resubmits. Show the review clock resetting. This closes the loop on the review workflow and demonstrates the two-way communication the SOO describes.

Stage 8, Approval, Payment & Permit Issuance

Show the approval decision with explanation, online payment simulation (credit card and check), permit issuance, and automated notifications to the applicant and relevant public channels. The SOO is very specific that fees must be transparent, payments must be PCI-compliant, and issuance must trigger public communication.

Stage 9, Post-Permit: Inspection, Enforcement & Renewal

This is the biggest addition from an initial framing. Show at least one of the following:

1. Inspector viewing their assigned inspection on a mobile device and updating compliance status
2. A Stop Work Order or Notice of Infraction being issued with communication to the permit holder
3. A renewal/extension request with expiration alert

Even if only one of these is fully shown, the demo must signal that TOPS modernization supports the full lifecycle, not just front-end intake. The SOO dedicates significant space to SWOs, NOIs, revocations, and renewal/extension processes.

Stage 10, Management Dashboard & Reporting

1. Operations view: pending applications, review bottlenecks, SLA performance, assignments, revenue
2. Executive view: macro trends, approvals/denials, enforcement activity, geospatial permit density by ANC/Ward
3. AI insight panel: pattern-based flags (e.g., “Construction permits in Ward 6 averaging 14% over SLA this quarter”)
4. Configurable, scheduled, and exportable reports in PDF, CSV, and Excel formats
5. Drill-down capability from executive summary to individual permit records

4. Agentic AI Features to Build In

AI features should feel embedded in the workflow, not bolted on. Each feature below maps to specific SOO language on AI use cases.

AI Feature	Where It Appears in the Demo
Intake validation & missing-item detection	Stage 3, real-time during application entry
Permit type guidance	Stage 3, suggests correct permit based on project description
DCMR regulation surfacing	Stage 3, displays relevant code sections (Titles 6, 9, 16, 18, 24)
Application & document summarization	Stage 5, one-click reviewer brief generation
Permit de-confliction	Stage 6, flags GIS and permit conflicts on the map
Routing recommendation	Stage 5, AI suggests reviewer assignment based on workload and expertise
Plain-language status explanations	Stages 4 & 7, explains permit status to the applicant in plain English
Pattern-based management insights	Stage 10, bottleneck and compliance trend flags for leadership

5. What to Build vs. Mock for the First Demo

Build for Real	Mock or Simulate
Full applicant intake flow (Stages 2–4)	Live production payment gateway
Reviewer queue & electronic plan review	Full data migration logic
GIS map with conflict and asset overlay	Every permit type (show selector, focus on one)
All eight AI features above	Complete admin configuration suite
Payment simulation (not live processing)	Deep back-office audit/compliance workflows
Management dashboard with sample data	Release management and training modules (new permits)
RBAC role-switching across three roles	Full MFA/SSO integration (show prompt, note integration path)
Mobile-responsive inspection screen	Geospatial reports beyond sample ANC/Ward data

Production integrations, full migration logic, and deep compliance workflows matter in the final solution. For the demo, represent them through mocked integrations, sample audit logs, architecture notes, and a roadmap slide.

6. SOO Details the Demo Should Not Miss

These are elements from the SOO that are easy to overlook but will resonate strongly with DDOT evaluators:

1. — Pre-application touchpoint

The SOO’s service delivery model starts before an application is submitted. Even a “Pre-Application Consultation” entry point or Early Work Review banner signals that we read the full document.

2. — Configurable review timeframes and visible SLA clock

The SOO is explicit that review periods must be configurable and clearly communicated to applicants, technicians, reviewers, and the public. A visible clock ticking in the reviewer queue and applicant dashboard is both technically grounded and visually compelling.

3. — DCMR regulation library

The SOO mentions maintaining a library of applicable DCMR regulations (Titles 6, 9, 16, 18, 24). Surfacing a relevant regulation citation during intake, even as a tooltip or sidebar, demonstrates DDOT-specific domain knowledge.

4. — Breadth of roles in the RBAC architecture

The SOO lists over a dozen roles beyond the core five: PDRM Coordinator, Plan Reviewer, Public Space Committee, NOI Administrator, Cash Management Officer, Confidential Reviewer, and others. The demo doesn't need to show all of them, but the role management screen should hint at this full roster.

5. — Asset management engine tied to GIS

The SOO explicitly calls for tracking and managing permitting assets like sidewalks and streets. Labeling an overlay in the GIS conflict view as “Asset Management” with a sidewalk or street status indicator costs almost nothing to add and shows that we absorbed the full SOO.

6. — Public notification system

The SOO specifies that DDOT must be able to communicate with the public about permit-related activities, system changes, new permit types, and after-hours permitting. A simple “Public Notifications” banner or admin panel screen captures this requirement.

7. Full Requirements

Demo Objective

Build a clickable/demoable tool that shows how a modern cloud-based permitting platform can improve applicant experience, reduce manual work, accelerate review and approval, strengthen coordination, support map-based permitting, handle payments and notifications, and extend through inspection/compliance and reporting. The experience must feel clearly tailored to DDOT public-space permitting, not a generic AI product.

Required Demo Capabilities

1. Applicant intake with account, project, and business context
2. Permit type selection and guided application flow
3. GIS-based location selection and conflict/asset/context visualization
4. Document upload and electronic plan review with redline comments
5. Reviewer work queue with routing, configurable review clocks, and status updates
6. Applicant response and resubmission flow
7. Fee estimate, payment simulation, and permit issuance
8. Public/user notifications and real-time status transparency
9. Inspection/compliance or renewal/enforcement follow-up (at least one post-permit screen)
10. Dashboard and reporting views for operations and executive leadership
11. Security cues: RBAC role-switching, MFA prompt, audit log sample, SSO placeholder
12. Mobile-friendly inspector view, multilingual toggle, and 508-aware design signals

AI Features to Wire In

1. Intake validation and missing-item detection (real-time during application)

2. Permit type guidance based on project description
3. DCMR regulation surfacing (relevant code citations during intake)
4. Application and document summarization for reviewers (one-click brief)
5. Permit de-confliction support using permit and GIS context
6. Reviewer routing recommendations based on workload and expertise
7. Plain-language status explanations for applicants and staff
8. Pattern-based management insights on bottlenecks, approvals, compliance, and workload

Design Requirements

1. Mobile-responsive, inspectors use the system in the field (SOO is explicit about this)
2. 508-compliant visual cues throughout: contrast, screen reader labels, keyboard navigation
3. Multilingual toggle visible even if not fully implemented
4. DDOT branding and DC public-space visual identity
5. Security indicators visible throughout: MFA prompt, RBAC role labels, audit log reference

Success Criteria

1. Shows a realistic DDOT permit journey end to end
2. Demonstrates clear improvement in transparency, efficiency, and coordination over the current TOPS system
3. Makes AI feel practical, embedded, and trustworthy, not the centerpiece
4. Reflects both applicant experience and DDOT internal operations
5. Signals that the platform supports the full permit lifecycle, not just intake
6. Makes it easy for DDOT to see this as the foundation for TOPS modernization
7. Shows DDOT-specific domain knowledge: DCMR, ANC/Ward reporting, permit types, role structure

8. What to Leave Out of the First Demo

For the first pass, do not try to fully implement the following. These matter in the final solution but should be represented through mocked integrations, sample logs, architecture notes, and a roadmap slide:

1. Every permit type (show the selector; build one)
2. Full production payment and SSO integrations
3. Full data migration logic from legacy TOPS
4. Complete admin configuration suite
5. Deep back-office audit and compliance workflows
6. Full release management or user training modules

Appendix: Key SOO Reference Details for John

This appendix pulls the DDOT-specific details from the Statement of Objectives that are most relevant to building a convincing demo. Use these as the source of truth for labels, names, and logic throughout the prototype, do not invent generic placeholders when the SOO provides the real ones.

A1, Permit Types (Use These Exact Names)

The SOO names five permit categories. The demo should use all five in the permit type selector, even if only one is fully built out:

1. Parking / Occupancy
2. Construction / Excavation
3. Commercial Vehicle
4. Public Space Rental / Annual
5. Special / Heritage Tree Permit

Private users of public spaces include homeowners, tenants, developers, government agencies (local and federal), utility companies, and businesses needing to occupy metered or unmetered curbside parking, sidewalks, alleys, or travel lane areas for activity or construction-related work.

A2, Full RBAC Role List (Design the Architecture for All of These)

The SOO defines five primary roles in detail and lists over a dozen additional roles. The demo only needs to show three roles in depth, but the role management and user provisioning screens should visually signal that this full roster is supported:

Role	Key Permissions per SOO
Applicant	Submit and track permit applications, view status, upload supporting documentation
Reviewer	Review applications, provide redline comments, recommend approval or denial
Technician	Approve or deny applications, issue permits, update permit conditions
Administrator	Manage user accounts, configure system settings, run reports on system activity
Auditor	Access audit logs, generate reports on user activity, perform compliance checks
Inspector	View assigned inspections, update status, access inspection history, issue citations, close permits
Manager / Chief / Supervisor	Operational oversight, workload management, reporting access
Cash Management Officer	Fee recording, payment verification, revenue reporting
Plan Reviewer	Detailed plan review, redline markup, and technical compliance assessment
Public Space Committee	Committee-level review and approval for designated permit types

Role	Key Permissions per SOO
PDRM Coordinator / Manager	Public space and design review coordination and management
NOI Administrator	Notice of Infraction administration and enforcement workflow
DDOT Contractor / Contract Manager	Contractor-facing access with scoped permissions
Confidential Manager / Reviewer / Technician	Scoped access for classified or sensitive permit data
Customer Experience & Relations Assistant	Applicant support, inquiry handling, status communication
Institutional Coordinator	Coordination for institutional applicants such as universities or hospitals
Visitor	Read-only public access to permitted activities and status

A3, Workflow Automation Requirements

The SOO specifies the following workflow automation behaviors. These should drive the logic of the routing, notification, and status update screens:

1. Automatically route permit applications to the appropriate reviewers based on permit type and project complexity
2. Send real-time notifications to permit applicants at each stage transition (submission, review, return, approval, denial, issuance)
3. Update permit application status in real time across all user views
4. Configurable review timeframes based on permit type and project complexity, these must be clearly communicated to applicants, technicians, reviewers, and the public
5. Automatically notify permit holders of upcoming permit expirations and trigger the renewal/extension workflow
6. Provide a clear process for SWOs, NOIs, and permit revocations including communication to the permit holder, an opportunity for input, and a defined timeline
7. Automate data entry and validation to reduce manual staff effort
8. Support de-conflicting permits using AI and GIS context before submission is completed

A4, Named Integration Targets (Use These in the Architecture View)

The SOO names specific systems that TOPS must integrate with. Use these names in the demo where mocked integrations are shown, they will resonate with DDOT staff who know these systems:

System	Integration Purpose per SOO
DC Master Address Repository	Address auto-population and validation during permit location selection
Payment Gateway (PCI-compliant)	Online fee payment via credit card, check, or other methods; in-person option at OCFO
Office of the Chief Financial Officer (OCFO)	In-person payment processing and financial reconciliation

System	Integration Purpose per SOO
Mapping / ArcGIS Service	Location selection, conflict checking, ANC/Ward/block tie-in, asset visualization
Public Utilities Systems	Cross-checking permit applications against road closures, utility work, and construction activity
Inspection & Compliance System	Sharing inspection results, compliance status, and enforcement data with applicants and management
DDOT CSR System	Connecting permitted activities to customer service requests and field reports
DDOT Asset Management System	Tracking permitting assets such as sidewalks and streets for conflict and condition awareness
Data Analytics Application	Permit usage pattern analysis, approval rate tracking, process improvement identification
Office of Judicial Hearings	Hearing scheduling and management for contested permits or enforcement actions
Transcriber Services	Hearing transcription integration
District Government Databases	Broader data sharing across DC government systems

A5, Required Report Types (Use These Names in the Dashboard)

The SOO specifies named report types. Use these exact names in the reporting and dashboard screens rather than generic labels:

1. Permit Application Status Reports, number of applications received, pending fees, pending assignment, assigned, withdrawn, lapsed, expired, surrendered, approved, and denied
2. Permit Issuance Reports, number and type of permits issued including duration and conditions
3. Revenue Reports, revenue from permits, fees, fines, deposits, and refunds
4. Enforcement Activity Reports, SWOs issued, NOIs, revocations, and compliance status
5. Review Cycle Reports, time in each review stage, SLA performance, bottlenecks
6. Geospatial Reports, permit frequency by ANC, Ward, block, and time of day

Report formats required by the SOO: PDF, CSV, and Excel. Reports must be configurable, schedulable (daily, weekly, monthly, annually), and exportable. Executive-level views should be macro summaries; management views should support drill-down to individual permit records.

A6, Service Delivery Model (The Full Lifecycle the Demo Must Cover)

The SOO defines the TOPS service delivery model as follows. This is the lifecycle the demo must represent end to end:

1. — includes Early Work Review and Non-permit inspections. Entry point before a formal application is submitted. Pre-application process
2. — guided intake, document upload, fee estimation, and electronic plan submission. Application submission

3. — routing, reviewer queue, comment and redline workflow, configurable review clocks, applicant response and resubmission, and final decision with explanation. Review and approval or denial
4. — payment processing, permit generation, and notifications to the applicant and the public. Permit issuance
5. — inspections, compliance monitoring, enforcement actions (SWO/NOI/revocation), renewals, and extensions. Post-permit services

SLA performance metrics the SOO specifies: application processing time, approval rate, compliance rate, and customer satisfaction. These should be visible in the management dashboard.

A7, UX, Accessibility & Communication Requirements

These SOO requirements should be visibly reflected in the demo design, not just noted in a future-state slide:

1. 508 compliance, high contrast mode, screen reader compatibility, and keyboard navigation must be present or clearly signaled
2. Mobile responsiveness, explicitly required for inspectors and field personnel accessing the system on the go
3. Multilingual support, permit applications must be completable in the user's preferred language; show a language selector
4. Simplified information presentation, the SOO explicitly calls out that the current TOPS overwhelms users; the demo UI should demonstrate clear, uncluttered, task-focused design
5. Account and business account management, users must be able to create business accounts, associate applications, designate projects, and assign or remove applicants from projects
6. Public notification capability, DDOT must be able to communicate with the public about system changes, new permit types, scheduled maintenance, new process announcements, and any permit-related information
7. Clear communication throughout, status updates, review decisions, and fee information must be communicated in plain language at every stage

Bottom Line

The combination of GIS conflict visualization and embedded agentic AI is the strongest differentiator in this demo. That pairing is specifically called out in the SOO and is exactly what distinguishes TOPS 2.0 from a basic forms upgrade. If we build the demo described here, we will show DDOT something they can actually imagine using, and that is the whole game.